

ADIT RAHMAN

[linkedin.com/in/adit-rahman](https://www.linkedin.com/in/adit-rahman) | arahmana@uwaterloo.ca | 613-263-2672 | github.com/adit-rahman

EDUCATION

University of Waterloo

Waterloo, Ontario

Bachelor of Computer Science, Co-op | GPA: 3.86/4.00

Sept. 2024 – Apr. 2029

- **Relevant Coursework:** Object-Oriented Software Development, Functional Programming (Advanced), Algorithm Design & Data Abstraction, Graph Theory, Logic and Computation

RELEVANT SKILLS

Languages: Python, TypeScript, C++, Ruby, JavaScript, Go, Java, SQL, HTML/CSS

Frameworks/Libraries: React, React Native, Node.js, Flask, BullMQ, Prisma, GraphQL, REST APIs, OpenTelemetry

Databases: PostgreSQL, MySQL, YugabyteDB, MongoDB, Redis, SQLite

Developer Tools: Docker, Kubernetes, CI/CD, Git, GitHub Actions, Buildkite, Prometheus, Grafana, Linux, Bash

PROFESSIONAL EXPERIENCE

Shopify

Jan. 2026 – Apr. 2026

Software Engineer Intern

Ottawa, Ontario

- Designed and shipped a production job system from scratch (Node.js, BullMQ, Redis, Kubernetes worker pods) with custom Prometheus metrics and OpenTelemetry instrumentation, processing **122K+ daily tasks**
- Migrated production database tables from Vitess (MySQL) to YugabyteDB for Shopify's ads domain, processing **11.59M writes** with a 100% success rate across a system serving **18M+ ads/month**
- Diagnosed a critical 130K-row data inconsistency during production migration, traced root cause to primary key duplication across 1.27M rows, and unblocked a cross-team data pipeline dependency within hours
- Built a staging environment from scratch across 5 repositories, independently configuring Docker and Kubernetes deployments, Buildkite CI/CD pipelines, secrets management, service authentication, and database provisioning

Shopify

Sept. 2025 – Dec. 2025

Software Engineer Intern

Ottawa, Ontario

- Re-architected merchant eligibility system with parallel validation for safe migration off an external dependency, achieving **p50 32ms** response latency and a **97.4% cache hit rate** across 10.2M production lookups
- Systematically decoupled the ads-publisher service from an external merchant model, refactoring GraphQL APIs, and service resolvers while introducing a new request abstraction to eliminate a hard architectural dependency
- Built centralized shop disablement logic handling **581 merchant lifecycle events with 0 errors**, consolidating previously fragmented merchant lifecycle management
- Optimized merchant earnings page by migrating data fetching to a client-side architecture, cutting page load times from **2-4s to ~500ms**

PROJECTS

Lock-In 🏠 | *Python, PyTorch, OpenCV, Streamlit, SQLite*

- Developed a **real-time focus detection system** classifying user engagement via webcam using a fine-tuned ResNet CNN, with a rolling probabilistic attention-scoring algorithm for context-aware feedback
- Architected a fully **on-device AI pipeline** combining PyTorch inference and OpenCV frame capture, ensuring user privacy with zero cloud dependency

Event Aggregator 📡 | *Go, React, TypeScript, SQLite, WebSocket, Docker*

- Built a **real-time event aggregator** in Go using goroutines and channels for concurrent data collection, with REST APIs for dynamic listener management without service restarts
- Designed a WebSocket streaming layer delivering live events to concurrent clients with minimal latency

Chess Engine 🏁 | *C++20, SDL2, WebAssembly*

- Developed a cross-platform chess engine in C++20 with SDL2 graphics rendering and a WebAssembly browser build, using Object-Oriented Programming and Observer pattern for modular multi-display architecture
- Implemented 5 AI difficulty levels with 4-ply Alpha-Beta pruning, piece-square evaluation, and quiescence search, cutting evaluated states by **95%** and move computation from 4s to **~200ms**